

EDUCATION

- **University of North Carolina at Charlotte** Charlotte, NC, USA
M.S. in Computer Science (Concentration: Artificial Intelligence) Aug 2025 – May 2027
 - GPA: 3.83/4.00
- **Eulji University** Seongnam, South Korea
Bachelor of Medical IT and Marketing (Minor in Silver Care Start-ups) Feb 2016 – Feb 2022
 - Relevant Coursework: Computer Algorithm, Medical Image Processing, Data Structure, Software Engineering, Practice of Data Analysis, Database, Mathematics
 - Awards: Department Highest Academic Scholarship (2021), Academic Excellence Scholarship (2021), Excellence in Club Activities Award (2021)

RESEARCH INTERESTS

- Large Language Models, Natural Language Processing, Retrieval-Augmented Generation (RAG), GraphRAG, Privacy Risks in AI Systems, Ontology-based AI

RESEARCH EXPERIENCES

- **Privacy Risks in AI Systems / LLM-based Applications** Charlotte, NC, USA
Graduate Researcher (Advisor: Prof. Liyue Fan) Jan 2026 – Present
 - Conducting ongoing research on privacy risks in LLMs, focusing on membership inference attacks and how sensitive information may be exposed, inferred, or misused in black-box attack settings.
 - Reviewing related literature on privacy, data leakage, retrieval-augmented generation, and security risks in AI-driven systems.
 - Exploring evaluation methods to analyze how document-based AI systems handle sensitive information during retrieval and response generation, including in-context probing and group-level information attacks.
- **From Fine-Tuning to Building Language Models for Different Domains** Charlotte, NC, USA
Graduate Researcher (Advisor: Prof. Mohsen Dorodchi) Dec 2025 – Present
 - Designing a Socratic-questioning-based educational chatbot to help students study more effectively, using RAG, PEFT/LoRA, and domain-specific knowledge integration.
 - Building a document-grounded chatbot that generates answers based on course materials and instructions, using Python-based frameworks and the ChatGPT API.
 - Studying how LLMs can support Socratic questioning, domain adaptation, and deployable language-model applications for education and professional training.
- **A Study on Automatic Reporting of Vehicle Accident Sensors** Seongnam, South Korea
Undergraduate Researcher (Advisor: Prof. Ki-Young Lee) Oct 2019 – Nov 2020
 - Developed an MCU-based sensor integration system detecting severe collisions via temperature, humidity, GPS, and Wi-Fi analysis to enable automated reporting to the nearest fire station.
 - Aimed to reduce fatalities and injuries through faster emergency intervention in driver-incapacitated collisions.

WORK EXPERIENCES

- **Electric Power Research Institute (EPRI)** Charlotte, NC, USA
Data Scientist Intern *May 2026 – Aug 2026*
 - Built a benchmark framework to evaluate document-grounded chatbot responses using RAGAS, focusing on answer faithfulness, relevance, and retrieval quality.
 - Analyzed how retrieved chunks, word relationships, and document context influence RAG-based chatbot answers in domain-specific energy-sector applications.
 - Worked with Retrieval-Augmented Generation (RAG), GraphRAG, Neo4j, and internal chatbot systems to support AI-driven knowledge retrieval and evaluation.
- **GEOTWO Co., Ltd.** Gyeonggi, South Korea
Full-stack Developer *Jul 2022 – May 2023*
 - Designed and developed web applications integrating front-end and back-end systems for performant, user-friendly solutions.
 - Collaborated with cross-functional teams to align technical implementation with project goals and user requirements.
 - **Key Project 1: QA Web Application** *Sep 2022 – Apr 2023*
Developed front-end features using React, TypeScript, jQuery, and Semantic UI; supported back-end development with Java/Spring Boot; used SVN for version control and team collaboration.
 - **Key Project 2: Reference Web Tutorial** *Jul 2022 – Aug 2022*
Built tutorials for O2Platform, O2Map, and O2asis; improved JavaScript async operations, code organization, and map-based functionality using OpenLayers.

MACHINE LEARNING & AI PROJECTS

- **House Price Prediction with Regression Models** Charlotte, NC, USA
Individual Project *Sep 2025 – Dec 2025*
 - Built an end-to-end regression pipeline on Kaggle’s “House Prices – Advanced Regression Techniques” dataset with 1,460 homes and 79 features using Python, Jupyter Notebook, pandas, and scikit-learn.
 - Implemented seven models, including Linear Regression, Ridge Regression, Polynomial Ridge Regression, Random Forest, Gradient Boosting, XGBoost, and LightGBM, using a unified preprocessing pipeline.
 - Achieved best performance with XGBoost ($R^2 \approx 0.92$, RMSE $\approx 24.5K$, and approximately 71% of predictions within 10% of true prices), outperforming classical baseline models.

PROFESSIONAL DEVELOPMENT

- **Java Android Web App Developer Program, Ezen IT Academy** Seoul, South Korea
Trainee *Dec 2021 – Jun 2022*
 - Completed a seven-month project-based training program on Open API utilization, Java, Android, and web application development.
 - Built foundations in object-oriented programming with Java and strengthened knowledge of data structures and algorithms.
 - Gained full-stack development experience with Vue.js, Spring Framework, MySQL, and AWS EC2.
 - **Key Project 1: Hey, Buddy (Collaboration Tool)** *Apr 2022 – Jun 2022*
Led a team-based full-stack project using Spring, MyBatis, and Tomcat 9.0; deployed on AWS EC2; implemented real-time video conferencing with WebRTC and Zoom API.
 - **Key Project 2: Today’s House Admin BBS** *Dec 2021 – Mar 2022*
Built full-stack features with Spring, Bootstrap, and MySQL, including authentication, file uploads, validation, map integration, geolocation, and social login APIs.

EXTRACURRICULAR ACTIVITIES

- **CAIR AI Research Club, University of North Carolina at Charlotte** Charlotte, NC, USA
Regular Member Sep 2025 – Dec 2025
 - Implemented a basic neural network using PyTorch, practicing data scaling, training loops, prediction generation, and foundational deep learning workflows.
- **Coding Club, Eulji University** Seongnam, South Korea
Regular Member Mar 2021 – Feb 2022
 - Led development of a digital currency mining game in C#, modernizing Pac-Man-style gameplay with updated aesthetics, planning, and team collaboration.

SKILLS

- AI & Machine Learning: Supervised regression, tree-based ensemble methods (Random Forest, Gradient Boosting, XGBoost), model evaluation (R^2 , RMSE, MAE)
- Programming: Python (primary), Java, JavaScript, React
- Data & Databases: pandas, NumPy, SQL (MySQL, PostgreSQL, Oracle)
- Tools & Platforms: Jupyter Notebook, scikit-learn, Git/GitHub, AWS, Spring Boot
- Visualization: Tableau, Matplotlib, Seaborn, Office365